

Assessing a Convection-Permitting Numerical Weather Model in Cities

Atmospheric numerical modeling in cities is critical to understanding the impacts of climate change in the world's most populous places—urban areas. Cities are a major source of anthropogenic greenhouse gas emissions (Wang et al., 2023), which further highlights the importance of accurate numerical modeling in urban areas to enhance climate-informed policy making. Urban areas are of exclusive interest to researchers due to the influence man-made structures exhibit on the adjacent atmosphere. For example, the urban heat island (UHI) effect is exclusive to cities (Oke et al., 2017). However, numerical weather models struggle to accurately resolve complex topographic features in cities, and consequentially, struggle to simulate the atmospheric conditions in these areas. Accurate, high-resolution numerical weather models are critical tools in a changing climate, especially when millions are at risk of its impacts.

I want to forge future advances in numerical modeling. In November 2023, this was an aspiration that enabled me to begin conducting a continental-scale model assessment of CONUS404, a 4-km convection permitting numerical weather simulation. Continental-scale model assessments of numerical weather models are exceedingly rare as studies typically constrain their assessments to specific regions or cities (Thompson et al., under review). CONUS404 had no comprehensive, continental-scale assessments, specifically against hourly, station-level observations in urban areas. Thus, I identified a knowledge gap in our pre-existing literature, and it was a question I wanted to answer to enhance our understanding of CONUS404's accuracy.

My model assessment needed to answer two questions: a) what are CONUS404's errors and biases in urban areas, and b) what is causing them? I utilized temperature, dewpoint, and wind speed in my model evaluation. These are key atmospheric variables that undoubtedly influence human comfort and are well-observed across the contiguous United States (CONUS). My project demanded a strong understanding of Python, the ability to visualize complex data structures, a demonstrated mastery in statistics, and an obligation to engage in multilayered problem solving. I had to learn, master, and apply these skills as I was simultaneously progressing towards my research objectives.

I utilized the National Center for Atmospheric Research's (NCAR) super-computers to access the CONUS404 research data archive. I needed to visualize CONUS404 not as a dataset, but as a map with thousands of unique indices. These indices stored the simulated temperature, dewpoint, and wind speed. I then built a Python script that took the latitude and longitude of the observation site and paired it with the nearest model indices. After I automated the pairing process, I retrieved 10-years of hourly, simulated data at 550 sites from across the CONUS. The sites for this evaluation were intentionally identified based upon several criteria, including their proximity to a major city and whether a site could adequately represent a rural area. Of the 550 sites, 208 sites were in urban areas and 342 were in non-urban areas.

I wanted to gauge the model's skill at regional- and continental-scales and at several temporal scales. So, I developed my research methodology to include spatio-temporal mapping, annual, decadal, and seasonal assessments, and a rarely utilized sub-daily diurnal assessment that divides each day into four six-hour segments. The

results of my assessment provided NCAR critical insights into CONUS404's performance in urban areas, which are now supporting its continued development.

I mapped a suite of statistical tests, including the coefficient of determination (R^2), mean absolute error (MAE), root mean squared error (RMSE), and mean bias error (MBE) for temperature, dewpoint, and wind speed. I also quantified the regionality of the modeled errors by dividing the CONUS into respective climates, which achieved my goal of understanding CONUS404's performance in different regional climates.

Contrary to my original hypothesis that I would find modeled cold biases in urban areas, I found that urban areas, overall, displayed warm biases. Upon conducting a seasonal analysis, I found CONUS404 indeed simulated cold biases in urban areas, but they were seasonally constrained (fig. 1d). My seasonal analysis found that winter was the worst performing season for all three atmospheric variables. These insights are critical because winter presents several complex atmospheric phenomena that have historically been difficult to simulate. However, of all my results, I was most surprised to find that CONUS404 was systematically underestimating wind speed (fig. 2c), especially as prior work has found that models tend to overestimate it. My project mentor and I communicated the wind speed underestimation to our collaborators at NCAR, and I have slated this area for future research and collaboration with CONUS404 model research and development scientists outside the lab. In the meantime, I needed to determine why urban areas, overall, exhibited temperature warm biases (fig. 2a) and a systematic dewpoint dry bias (fig. 2b).

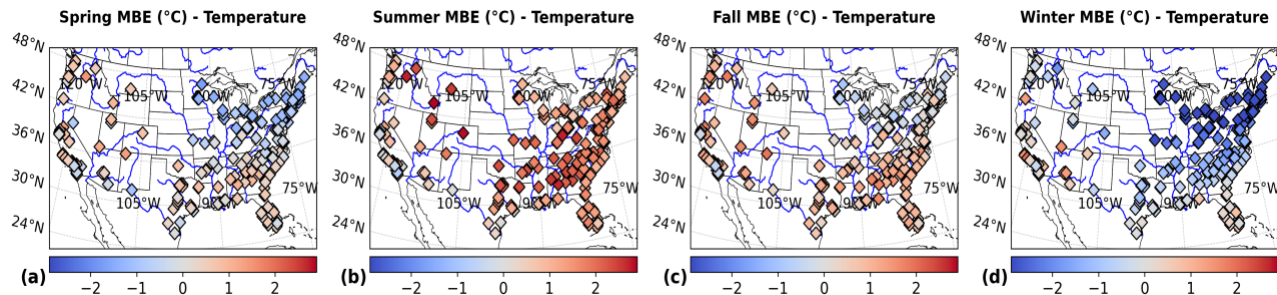


Fig 1: Temporal and spatial MBE map of temperature. Temperature shows strong summertime warm biases in urban areas that transition to wintertime cold biases, especially in the northeast U.S.

Testing the dependency of CONUS404's MBE on urban fraction, a measure that determines the fraction of "built-up" area in a model grid, helped explain my findings. As a result, fig. 2 demonstrates key weaknesses in using the bulk urban parameterization to represent cities in numerical weather models. This scheme characterizes cities as flat surfaces with increased friction and no vegetation.

Modeled temperature MBE (fig. 2a) showed a multilinear relationship on urban fraction. In the bulk urban parameterization, areas of higher urban fraction have a lower amount of bare soil and vegetation (He et al., 2023). This is physically accurate in cities and will thus lead to a lower MBE when simulating temperature in high urban fraction areas. Lower urban fractions have fewer urban structures, which physically warm faster than surrounding rural areas, causing CONUS404 to simulate the largest warm biases in the lowest urban fractions. Compare this to dewpoint (fig. 2b), which clearly shows an increasing dry bias with increasing urban fraction. Higher urban fractions have an underrepresentation of urban vegetation, which is leading to significant dry biases.

Wind speed MBE (fig. 2c) also showed a multilinear dependency on urban fraction. This shows that the wind speed biases can also be attributed to the bulk urban parameterization. CONUS404 is likely overestimating the friction, which is causing the wind speed underestimation. Moreover, I highlight an area of uncertainty in this hypothesis and propose that part of the underestimation can be explained by an observation bias. Meteorological observations tend to be recorded in fields—not within cities. Cities slow wind speeds, and it is possible CONUS404 is accurately simulating this real-world phenomenon. However, I found a systematic underestimation across all spatial and temporal scales tested. Without additional wind speed observations within cities, it is difficult to conclude whether the underestimation is a true model error.

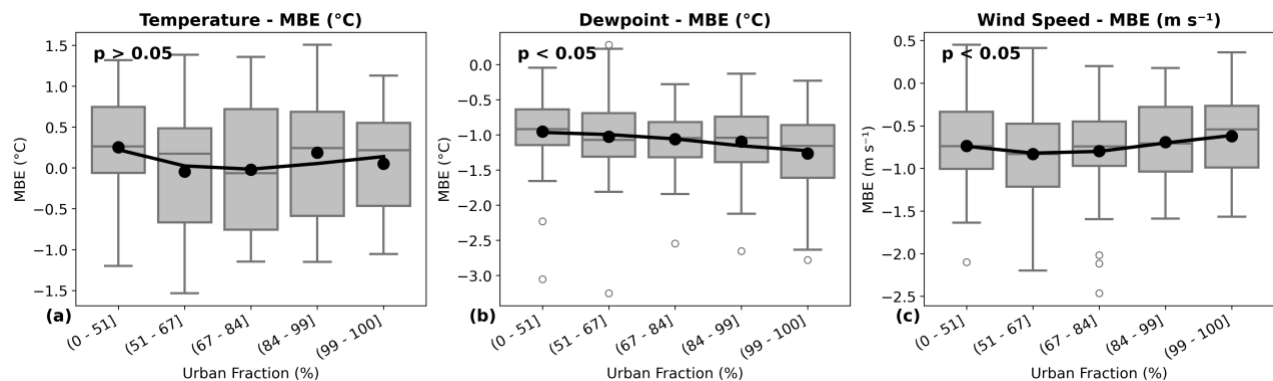


Fig 2: Temperature (a), dewpoint (b), and wind speed (c) MBE dependency on urban fraction. Dewpoint errors are nearly linear whereas temperature and wind speed show multilinearity. Particularly, modeled temperature and wind speed MBE peak between 51-67%.

My analyses demonstrate the inadequacy of utilizing the bulk urban parameterization to resolve key meteorological variables in a high-resolution, convection-permitting simulation at continental- and regional- scales in urban areas. As a result, my work is driving and informing the future model development of such state-of-the-art numerical weather models. Additionally, I am introducing a new standard for model assessment—comprehensive is at a continental scale. This project has motivated me to pursue a research career focused on developing and improving these numerical weather models, especially where they matter most—in the world’s most populous places, urban areas.

References

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